

3D Object Detection with LiDAR-Camera Fusion

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PROJECT 3 – 3D PERCEPTION

Goal: The project aims to develop a 3D Object Detection system based on deep learning, which leverages the multi-modal fusion of data acquired from LiDAR sensors and RGB cameras. The objective is to detect and localize three-dimensional objects (such as automobiles, pedestrians, and cyclists) within autonomous driving scenarios, utilizing the KITTI-360 dataset.

EXECUTION:

STEP 1: Download the KITTI-360 dataset from the official repository. For this project, a subset of the dataset was acquired, specifically selecting the relevant urban driving sequences. The downloaded data includes:

- Camera image files (.png)
- LiDAR point cloud files (.bin)
- Calibration files (.txt) containing intrinsic and extrinsic parameters for sensor alignment
- data_3d_bboxes files (.xml)

STEP 2: Network Architecture Design

For processing the point cloud data, we employed a LiDAR Backbone based on the PointPillars architecture. Concurrently, an Image Backbone utilizing ResNet50 was used for the camera data. To effectively combine the information from both modalities, an Attention-Based Fusion Module was implemented. Finally, a dedicated Detection Head module performed the final predictions.

Component Details:

- LiDAR Backbone (PointPillars):

- Rationale: The PointPillars architecture was selected due to its computational efficiency and favorable trade-off between accuracy and speed compared to alternative methods.
- Structure: Consists of 3 layers.
- Image Backbone (ResNet50):
 - Rationale: ResNet50 was chosen for its widespread use and its pre-trained ability to effectively recognize textures, shapes, and generic objects, providing robust feature extraction.
 - Structure: Comprises several layers, typically segmented into Low-level, Mid-level, and High-level feature maps.
 - Training: All layers are trainable, enabling precise fine-tuning on the KITTI-360 dataset.

FUSION MODULE

The fusion module performs the following critical operations:

- Spatial Alignment: Aligning features from the two distinct sensor coordinate systems.
- Concatenation: Combining the spatially aligned features.
- Attention Mechanism: Applying an attention mechanism to assign importance weights to features from each modality.
- Weighted Fusion: Generating the final fused feature representation based on the attention weights.

DETECTION HEAD

The detection head is responsible for generating the final 3D bounding box predictions, including:

- Classification: Predicting the object class (e.g., car, pedestrian).
 - *Loss Function*: Binary Cross Entropy standard
- Regression Head: Predicting the 3D bounding box parameters (size, location).
 - *Loss Function*: Smooth L1 Loss
- Direction Head: Predicting the object orientation/heading.
 - *Loss Function*: Cross Entropy Loss

STEP 3:

Hyperparameter Optimization and Training Setup

The final stage involved the adjustment and modification of various module parameters, with a particular focus on the parameters of the training loop, including: Epochs,

Batch Size, Train/val Spli, Gradient Clipping

CONCLUSIONS

For the project I used Google collab pro to have enough computing power to train the network, the results of the network training are not bad.

From the results obtained, it can be seen that the model works; the training loss and validation loss decrease.

However, they fail to fall below a certain threshold.

This is certainly due to the simplicity of the model, the low number of samples used, and the lack of perfect parameter settings.